

Detection limits of matrix-assisted laser desorption/ionisation mass spectrometry coupled to chromatography – a new application of solvent-free sample preparation

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The detection limits of matrix-assisted laser desorption/ionisation mass spectrometry (MALDI-MS) by semi-online coupling with chromatography were investigated using various mixtures of polyethylene oxides (PEOs) with different end groups. In contrast to the common ‘dried-droplet’ sample preparation technique, which results in an inhomogeneous sample-to-matrix ratio within the MALDI spot, the used coupling technique offers a very high reproducibility combined with surpassing sensitivity of a few femtograms over a broad range of sample-to-matrix ratios. These results are in correlation with the results of the recently established solvent-free MALDI-TOFMS method utilising the grinding approach and are also of assistance towards the more theoretical aspect of MALDI that suggests that there is no necessity for an analyte incorporation into a matrix crystal for excellent matrix-assistance. Copyright © 2005 John Wiley & Sons, Ltd.

Recently, the principle of solvent-free sample preparation was introduced by Trimpin *et al.*^{1–3} This involves the mixing of sample and matrix components by intense grinding in a mortar or, more efficiently, in a mill without any solvent. Primarily developed for the matrix-assisted laser desorption/ionisation mass spectrometric (MALDI-MS) investigation of insoluble samples this method has become increasingly useful for the analysis of common synthetic and biopolymers.^{4–6} In contrast to solvent-based preparation techniques an intense grinding of samples and matrix results in a very homogeneous powder downward to micrometer scale. A segregation based on different solubility of the both compounds in the solvent can be avoided. This leads to a very high spot-to-spot reproducibility. Finally, the ratio of sample and matrix in the mixture seems to be less crucial for a good spectra quality.⁷

The latter reason in particular is extremely important for coupling chromatography with MALDI-MS. This method combination is very important for a MALDI analysis of broad distributed polymer samples and for a two-dimensional (2D) analysis of heterogeneous polymers.^{8–10} The sample is sprayed onto the MALDI target by means of an interface. The solvent is evaporated in a nitrogen stream using a heated needle. However, the sample concentration within the chromatographic peak is not constant, i.e. very low at the flanks as well as comparatively high at the top of the peak. In contrast to that, the matrix solution is sprayed with a constant concentration onto the target before the sample is deposited or simultaneously with the sample. Therefore, different sample-to-matrix ratios are obtained, which strongly differ

from optimal values. This drawback can be overcome by adapting the matrix concentration to the peak intensity (e.g. controlling of the matrix pump flow by the outlet signal of an additional concentration detector).

Additionally, the use of a **micro-high-performance liquid chromatography (μ-HPLC)** system should be investigated. On the one hand, applying such a system, the amount of solvent to be evaporated can be drastically reduced. On the other hand, the absolute amount of sample deposited onto the MALDI target strongly decreases too.

Our aim was to establish such conditions of sample deposition that can be compared to the ‘solvent-free’ sample preparation method. This could be achieved when the solvent is completely evaporated before the sample molecules impinge the matrix on the pre-coated target.

The influence of various matrix-to-sample ratios on the spectra quality and sensitivity using these optimised conditions was investigated.

EXPERIMENTAL

Materials

The polyethylene oxide samples (PEO 1000 and 6000) were obtained from Polymer Standards Service (PSS, Mainz, Germany). Functionalised PEO samples were synthesised and supplied by Dr. H. Budde (Martin-Luther-University of Halle-Wittenberg, Germany). The chromatographic solvents used were HPLC-grade. Solvents as well as matrices for MALDI-MS were obtained from Fluka (Germany). No additional purification was performed.

Chromatography

For chromatographic investigations two systems were used. The high-performance liquid chromatography/size-exclusion

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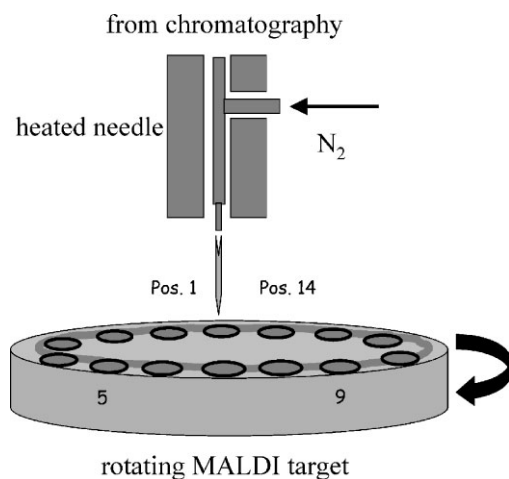


Figure 1. Principle of spraying onto a rotating MALDI-MS target.

chromatography (HPLC/SEC) investigations on the analytical scale (system 1) were performed by means of a HP 1090 chromatography system equipped with two C18 columns (250×4.6 mm and 150×4.6 mm i.d., YMC Europe GmbH). The system was coupled to an evaporative light-scattering detector (ELSD) (SEDEX 45, Sedere, France). For μ -HPLC investigations (system 2) an Agilent 1100 series μ -chromatography system equipped with two C18 columns (150×1 mm and 150×0.5 mm i.d., GROM ODS 120 and Zorbax SBC 18) was chosen. The injected sample volume was between 0.5 and $2 \mu\text{L}$ at a flow rate of $20 \mu\text{L}/\text{min}$. The SEDEX 75 ELSD (Sedere) was used for detection. The chromatographic systems were kept at a constant temperature of 45°C .

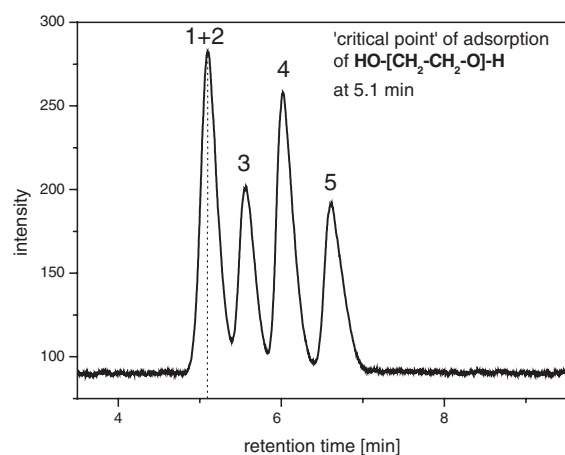


Figure 2. Chromatogram at 'critical point' of adsorption of a PEO mixture.

Size-exclusion chromatography (SEC mode, system 1)

In SEC mode, methanol (100%) was used as eluent. The flow rate was $0.5 \text{ mL}/\text{min}$.

Liquid adsorption chromatography at 'critical' conditions of adsorption (LACCC mode, system 2)

The determination of 'critical conditions' of adsorption was performed starting with methanol as a thermodynamically 'good' solvent. Successive addition of water as a thermodynamically 'poor' solvent resulted in a change of the chromatographic mode to adsorption mode. The 'critical conditions of adsorption' were established when samples with the same chemical composition, but different molecular weight, eluted in one peak. For the investigated polymer system the 'critical

No.	Average molecular weight Mw (Da)	Structure
1+2	1000 + 6000	$\text{H}-\left[\text{O}-\text{CH}_2-\text{CH}_2\right]_n-\text{OH}$
3	500	$\text{H}_3\text{C}-\left[\text{O}-\text{CH}_2-\text{CH}_2\right]_n-\text{OCH}_3$
4	1000	$\text{H}_2\text{C}=\text{C}(\text{CH}_3)-\text{C}(=\text{O})-\left[\text{O}-\text{CH}_2-\text{CH}_2\right]_n-\text{OCH}_3$
5	1000	$\text{H}_2\text{C}=\text{C}(\text{CH}_3)-\text{C}(=\text{O})-\left[\text{O}-\text{CH}_2-\text{CH}_2\right]_n-\text{O}-\text{C}(=\text{O})-\text{C}(\text{CH}_3)=\text{CH}_2$

Figure 3. Chemical structures, average molecular weights and end groups of investigated PEOs.

solvent composition' (CSC) could be obtained by adjusting a methanol/water mixture of 84:16 wt %. The composition of the mobile phase was adjusted by weighing of the solvents. The eluent flow was 20 μ L/min.

MALDI-MS

A Bruker Reflex III MALDI mass spectrometer (Bruker-Daltonik, Bremen, Germany) operating at 20 kV acceleration voltage was used. For ionisation/desorption a UV laser working at a wavelength of 337 nm was applied. The laser pulse length was 3 ns. Typically, 100–200 transients were accumulated for one spectra. The mass spectrometer was calibrated using different PEO standards. There was no additional salt added.

The preparation of MALDI spots according to the 'dried-droplet' method was performed using premixed solutions: 2 mg of solid matrix were dissolved in 1 mL tetrahydrofuran (THF). Two different matrices – trihydroxyanthracene (dithranol) and *trans*-2-(3-(4-*tert*-butylphenyl)-2-methyl-2-propenylidene)malonitrile (DCTB) – were used. A volume of 20 μ L of each matrix solution was mixed with 20 μ L of the polymer solution. For the preparation of spots, 1 μ L of the mixture was deposited on the target by means of a pipette. After evaporation of the solvent the target was inserted into the mass spectrometer.

Various solutions with different polymer concentrations were prepared.

Coupling of MALDI-MS with chromatography

A commercially available interface (LC 500, LabConnection, USA) was used for semi-online coupling of liquid chromatography and mass spectrometry.

In the first step, the target was coated with matrix by spraying using a secondary pump (Knauer, Germany). The principle of sample deposition is shown in Fig. 1. 1,8,9-Trihydroxyanthracene (dithranol), 2,4,6-trihydroxyacetophenone (THAP) and *trans*-2-(3-(4-*tert*-butylphenyl)-2-methyl-2-propenylidene)malonitrile (DCTB) dissolved in THF (ca. 1 mg/mL) were used. The deposition flow rate was 0.4 mL/min. The solvent was evaporated in a nitrogen gas stream under pressure at an elevated temperature of 130°C. The applied gas pressure was 20 psi which corresponds to a nitrogen gas flow rate of 2 L/min.

Afterwards, the eluate of a chromatographic run was sprayed onto the pre-coated target. In order to avoid crystallisation of matrix on the tip of the sprayer needle the temperature had to be carefully adjusted. Due to the amount of water in the solvent mixture of system 2 the evaporation temperature (150°C) as well as the gas stream (30 psi = 3 L/min) had to be increased to achieve a complete evaporation.

RESULTS AND DISCUSSION

Adjusting of 'critical' parameters of separation of PEOs

For the separation of PEOs many 'critical conditions' of separation are known. These conditions depend on the polarity of the used column as well as on the solvent system. For our experiments five PEOs with four different end groups

were investigated in a MeOH/H₂O system. As explained before, the 'critical mode' is reached when polymers with same end groups elute at identical retention times. Functionalised PEOs with different end groups, however, elute at different times. This is shown in Fig. 2. Consequently, the analysis of PEOs with four different end groups results in four peaks, whereas non-functionalised PEO 1000 and 6000 elute at the same time. The corresponding end group structures, confirmed by means of MALDI-MS, are shown in Fig. 3.

Choice of matrix

In MALDI-MS PEOs are known to be easily ionised with many matrices. Therefore, two commonly used compounds (dithranol, THAP) were compared to a relatively new matrix substance (DCTB) with regard to peak intensity and spectra quality. Because of their tendency to form large crystals in the solid state, 2,5-dihydroxybenzoic acid (DHB), respectively, due to cluster formation in the mass region around 1000 Da, hydroxyphenylazobenzoic acid (HABA) were not applied for

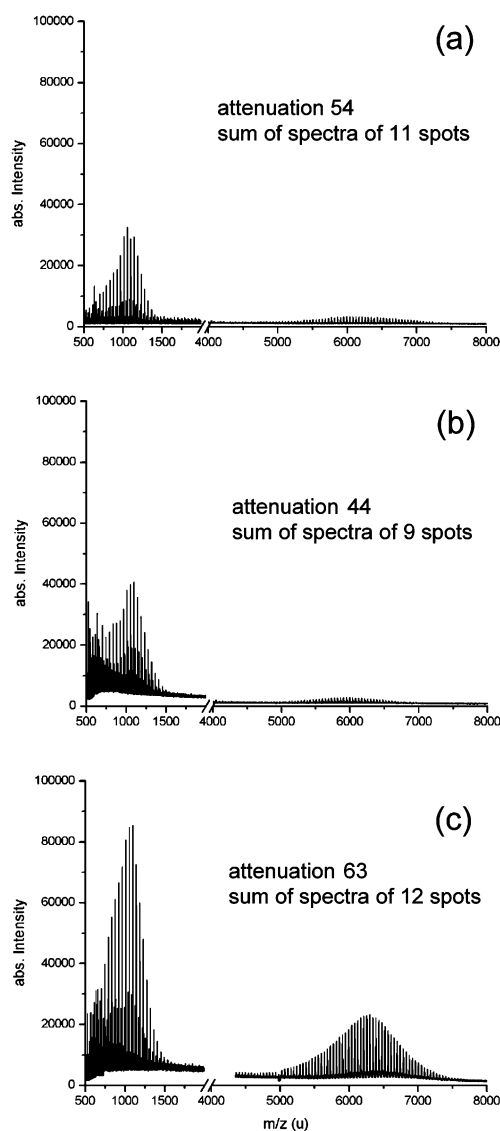


Figure 4. MALDI mass spectra of a 1:1 (w/w) mixture of PEO 1000 and 6000 using different matrix compounds: (a) dithranol, (b) THAP, and (c) DCTB.

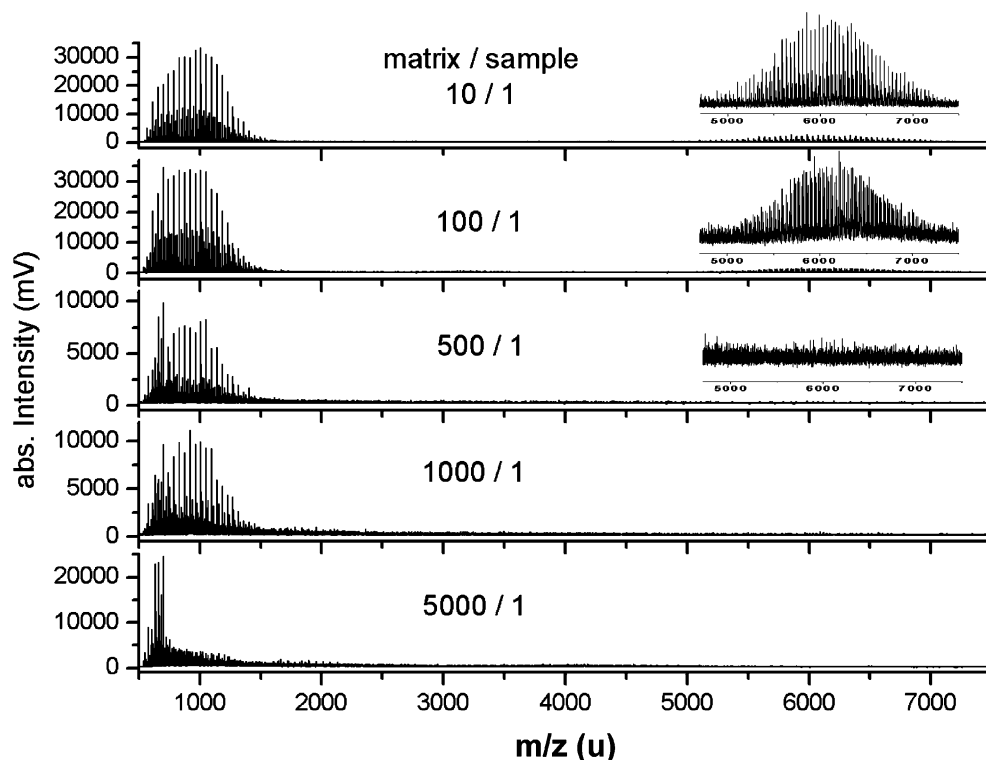


Figure 5. MALDI mass spectra of a 1:1 (w/w) mixture of PEO 1000 and 6000 (matrix: DCTB, preparation applying the 'dried-droplet' method).

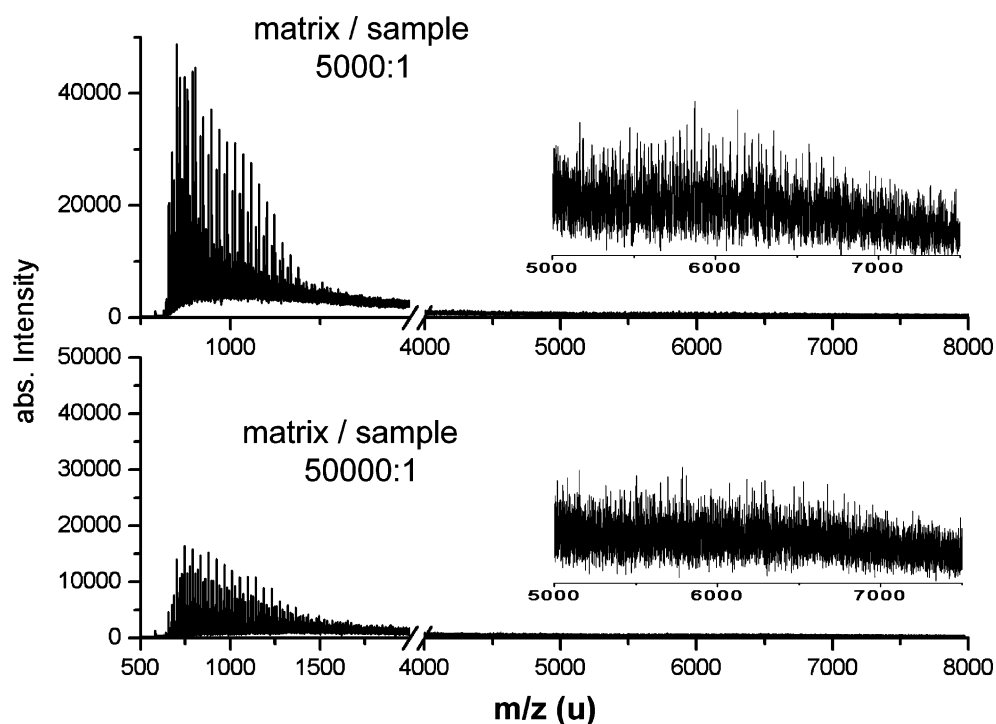


Figure 6. MALDI mass spectra of a 1:1 (w/w) mixture of PEO 1000 and 6000 (matrix: DCTB, the spraying method, spraying on one spot).

Table 1. Mixture of PEO 1000 and 6000 (1:1, w/w) used for spraying in SEC mode in μ -HPLC

Sample conc.	2.140 mg/mL		0.210 mg/mL		0.021 mg/mL	
Injection volume	2.0 μ L	0.5 μ L	2.0 μ L	0.5 μ L	2.0 μ L	0.5 μ L
Estimated amount per spot	50.0 ng	12.5 ng	5.0 ng	1.2 ng	500 pg	125 pg
Matrix/analyte ratio	500:1	2,000:1	5,000:1	20,000:1	50,000:1	200,000:1

these investigations. After separation in SEC mode (system 1, 100% MeOH) a 1:1 (w/w) mixture of PEO 1000 and 6000 was sprayed continuously onto the rotating MALDI target (see Fig. 1), which was pre-coated with matrix. Figure 4 shows the sum of MALDI spectra recorded by scanning of single target spots (in the centre of each spot, along the chromatographic direction, see Fig. 1). In each sum spectrum, the number of spots (where spectra could be accumulated) and the attenuation of the mass spectrometer are given. In contrast to dithranol (Fig. 4(a)) and THAP (Fig. 4(b)), the MALDI mass spectra obtained using DCTB as matrix offered the highest signal intensity recorded at lowest laser power (i.e. highest attenuation). Especially in the higher mass range of the spectrum a surpassing intensity of PEO 6000 Da peak signals was observed. Furthermore, spectra at 12 spots (of 14 possible spots on the target) could be recorded, that can be attributed to a higher sensitivity even at very low sample concentrations (i.e. at the slope of the chromatographic peak).

Therefore, DCTB was chosen for the following experiments without the addition of salt. Sodium and, with lower intensity, potassium adduct ions were detected.

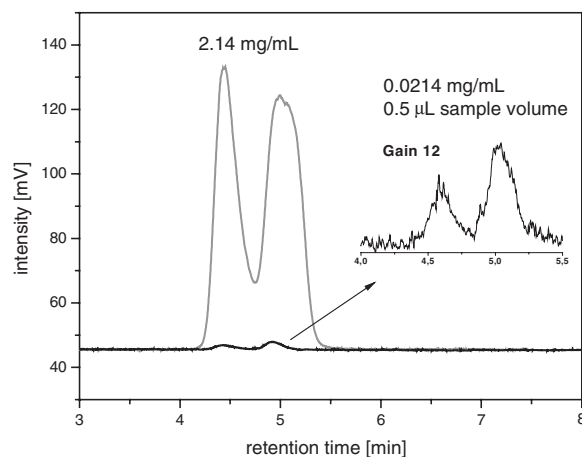


Figure 7. SEC chromatogram of a 1:1 (w/w) mixture of PEO 1000 and 6000 using two different concentrations (ELS detector: gain 7).

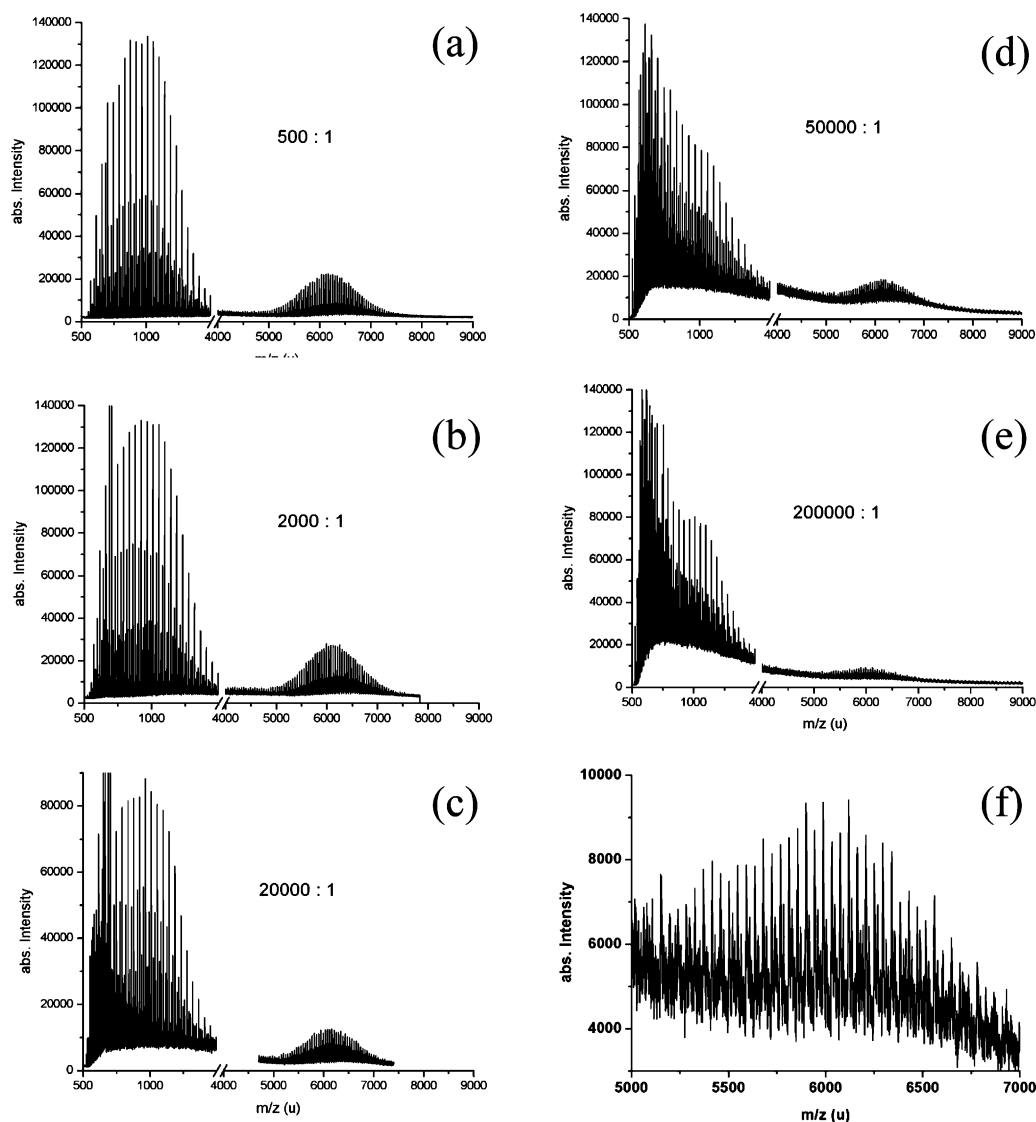


Figure 8. MALDI mass spectra of a 1:1 (w/w) mixture of PEO 1000 and 6000 using different sample/matrix ratios (a–e) and enlarged region at 6000 Da at a sample matrix ratio of 200 000:1 (matrix: DCTB, spraying on a rotating MALDI target).

Table 2. Mixture of five PEO samples (1–5) used for spraying in 'critical mode' in μ -HPLC

Sample conc.	2.570 mg/mL	0.257 mg/mL
Injection volume	0.5 μ L	0.5 μ L
Estimated amount per spot	6.0 ng	600 pg

Comparison of hand spotting ('dried-droplet' method) and spraying

For a comparison of the hand spotting and the spraying methods the above mentioned mixture of PEO 1000 and 6000 was used as well. Figure 5 shows MALDI mass spectra of matrix/sample mixtures with different ratios. A ratio of 10:1 (matrix/sample) results in spectra with high resolution for both polymers. However, just a moderate increase in the amount of matrix up to a ratio of 100:1 leads to deterioration of the spectra quality. Although the intensity of signals remains more or less unaffected, the signal-to-noise ratio in the PEO 6000 region is much lower. In contrast to the PEO 1000 signal, which can be detected up to a matrix/sample ratio of 1000:1, a further increase in the amount of matrix in the mixture causes a disappearance of PEO 6000 peaks. At a higher ratio of 5000:1 neither PEO 1000 nor PEO 6000 signals can be detected in sufficient quality.

Spraying was performed using the LC 500 interface. Single spots were created, i.e. the MALDI target did not rotate. The results are shown in Fig. 6. In contrast to the 'dried-droplet' method spectra of PEO 1000 and 6000 could be easily recorded using a mixture ratio of 5000:1. Surprisingly, even at a ratio of 50000:1 (matrix/sample), PEO 1000 signals could be detected. Obviously, the spraying method seems to be up to 100 times more effective in the 1000 Da range and 10 times more effective in higher mass regions.

Spraying by means of μ -HPLC (in SEC mode)

In order to obtain more knowledge about the applicability of μ -HPLC semi-online coupled to MALDI-MS, a PEO mixture (PEO 1000 and 6000) was sprayed in different concentrations on a rotating pre-coated MALDI target (Table 1). At the chromatographic settings (see Experimental section) the preparation of absolutely dry sample spots on the target at comparatively moderate evaporation conditions (100°C, 10 psi = 1 L/min N₂) was enabled. Figure 7 shows two chromatograms recorded before spraying. Two peaks were detected which can be attributed to PEO 1000 and 6000. The signal intensity at the lowest sample concentration of 0.0214 mg/mL is close to the detection limit of the detector. Nevertheless, as shown in Figs. 8(a)–8(e), and in more detail in Fig. 8(f), MALDI mass spectra could be recorded even at these low concentrations, which corresponds to an amount of 20 fmol of PEO 6000.

Spraying by means of μ -HPLC (in LACCC mode)

For the LACCC investigations the five-component PEO mixture used to adjust the 'critical' parameters (see Fig. 2) was sprayed onto the MALDI target in two different concentrations. The conditions are outlined in Table 2. At the lowest concentration an estimated amount of 600 pg is located at one MALDI spot. The corresponding mass spectra recorded

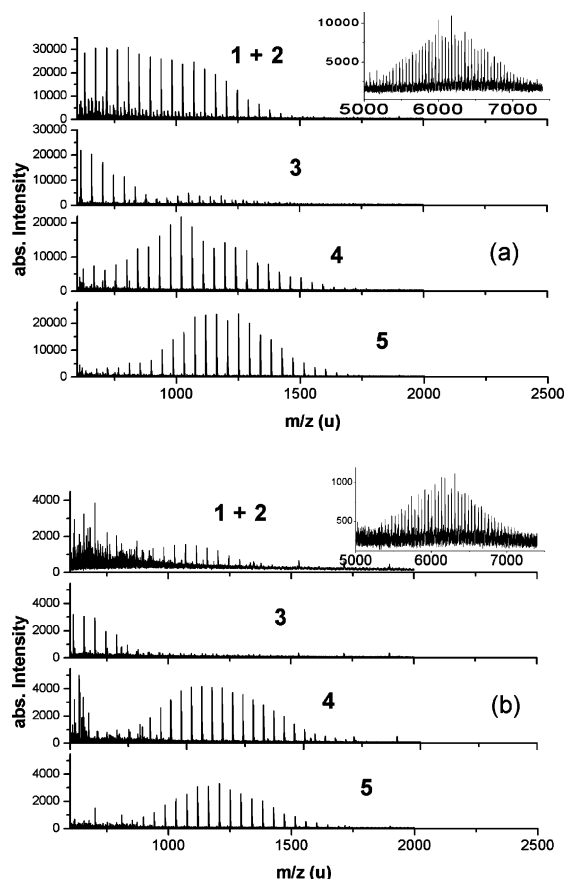


Figure 9. MALDI mass spectra of a mixture of five PEOs (structures shown in Fig. 2) after separation at 'critical point' of adsorption sprayed on a rotating MALDI target (matrix: DCTB, sample volume 0.5 μ L, sample concentration: (a) 2.570 mg/mL and (b) 0.257 mg/mL).

at MALDI target spots 2, 4, 6 and 8 (close to the maximum of chromatographic peaks shown in Fig. 2) are presented in Figs. 9(a) and 9(b). Very good spectra could be obtained using the higher sample concentration (Fig. 9(a)). However, even at the lower concentration (Fig. 9(b)), spectra were obtained which enabled a clear determination of each PEO functionality. Merely the intensity of signals of PEO 1000 (sample 1) is

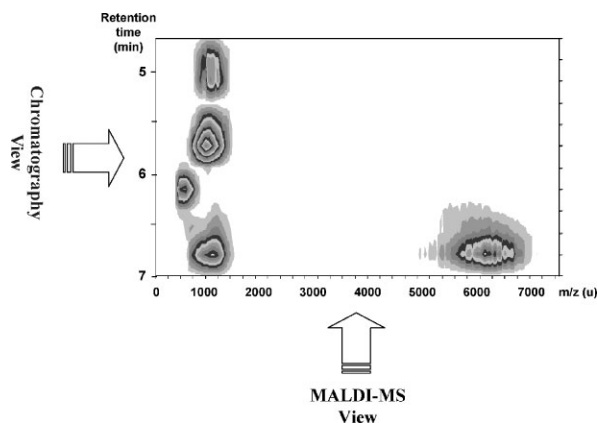


Figure 10. Two-dimensional plot combining twelve MALDI mass spectra and the chromatographic information recorded at 'critical point' of adsorption of a mixture of five PEOs.

very low. Due to non-baseline separation in chromatography, spectra exhibit less-intense additional signals which can be attributed to sodium and potassium adduct ions of PEOs with other end groups.

Due to different ionisation probabilities of polymers having the same structure but various end groups, one has to keep in mind that MALDI-MS cannot be regarded as a quantitative method. This disadvantage can be overcome by coupling MALDI with chromatography which is quantitative. Thus, the information from MALDI spectra 1–12 (partially shown in Fig. 9) were combined with the peak intensities from chromatography (see Fig. 2). This results in the 2D plot shown in Fig. 10. Since two polymers (PEO 1000 and 6000, 1:1 w/w mixture) are hidden in the first peak at 5.1 mL retention volume, its intensity used for calculation of the 2D contour plot intensity was halved. These plots can be used as a fast 'fingerprint' for the analysis of polymer mixtures.

CONCLUSIONS

Applying the spray sample deposition method very homogeneous analyte/matrix mixtures could be obtained. These spots show a high MS sensitivity at comparatively low laser power over a broad range of analyte-to-matrix ratios. Subsequently, our results were compared to results obtained with the conventional solvent-based preparation method ('dried-droplet' method). In contrast to solvent-based MALDI-MS,

the spray technique described in this paper offers a few important advantages, the most promising of them are improvements in compatibility of analyte and matrix, higher reproducibility of spectra due to better homogeneity and, in particular, a higher sensitivity. The spraying approach combines the advantages of chromatographic separation and solvent-free sample preparation for MALDI-MS. Overall, it can be concluded that omitting the solvent in the sample preparation-deposition step improves the MALDI result inherently.

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